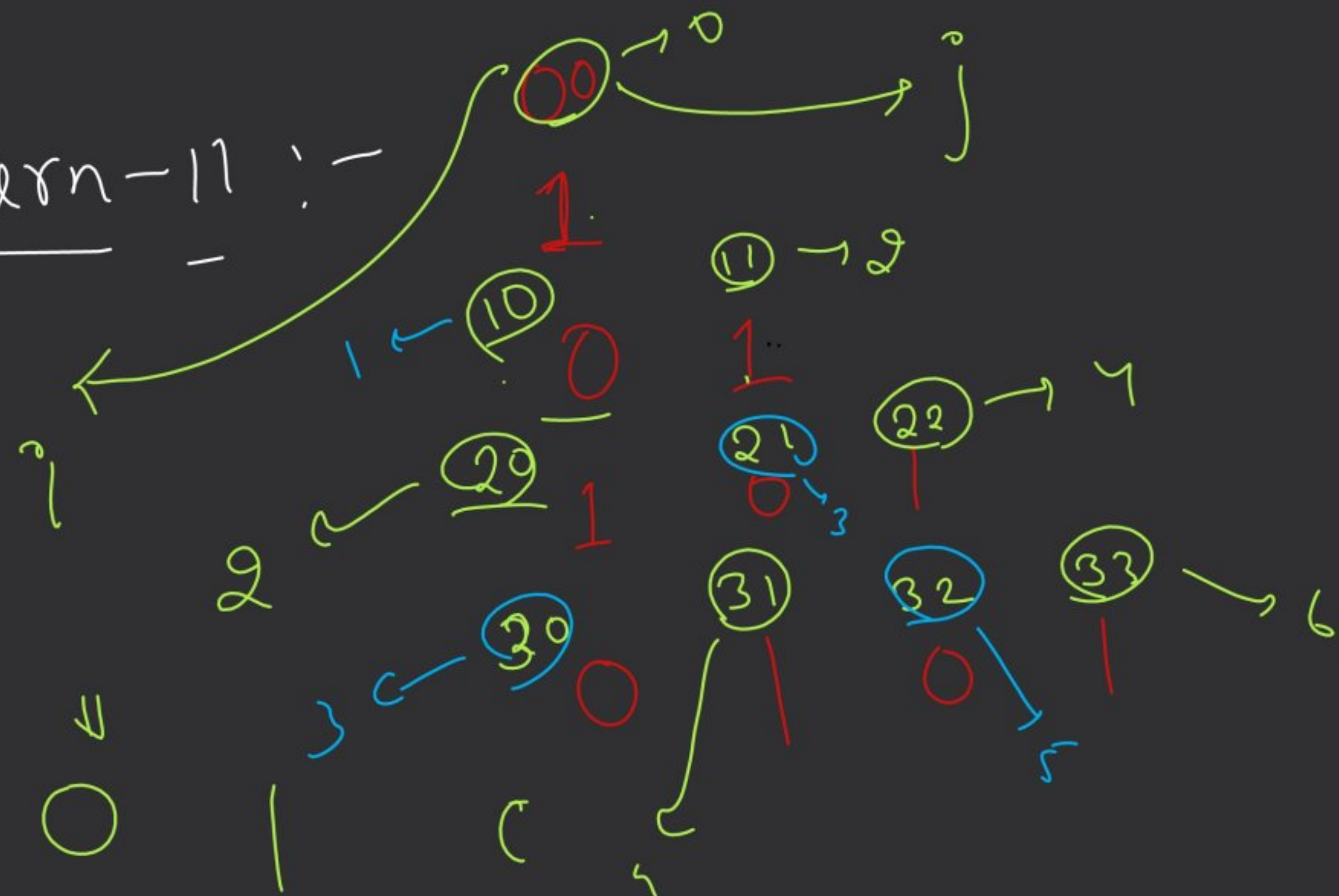
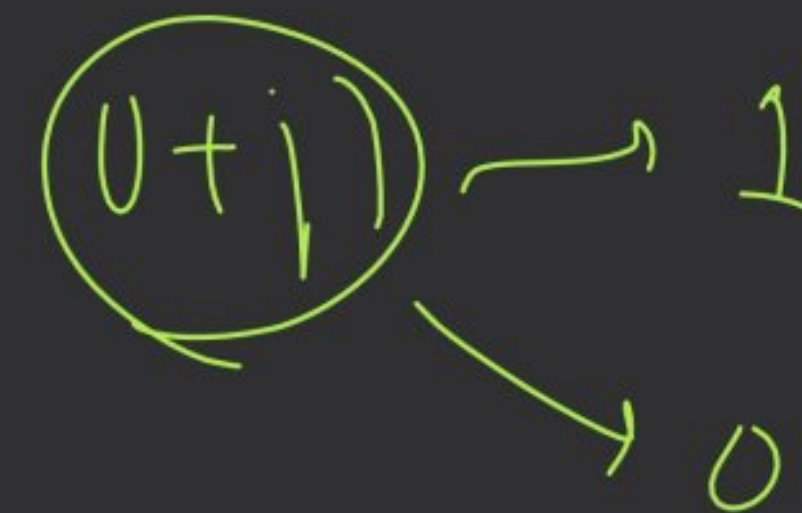


Pattern-11 :-



→ Loops

→ $j \Rightarrow (0 \rightarrow i)$



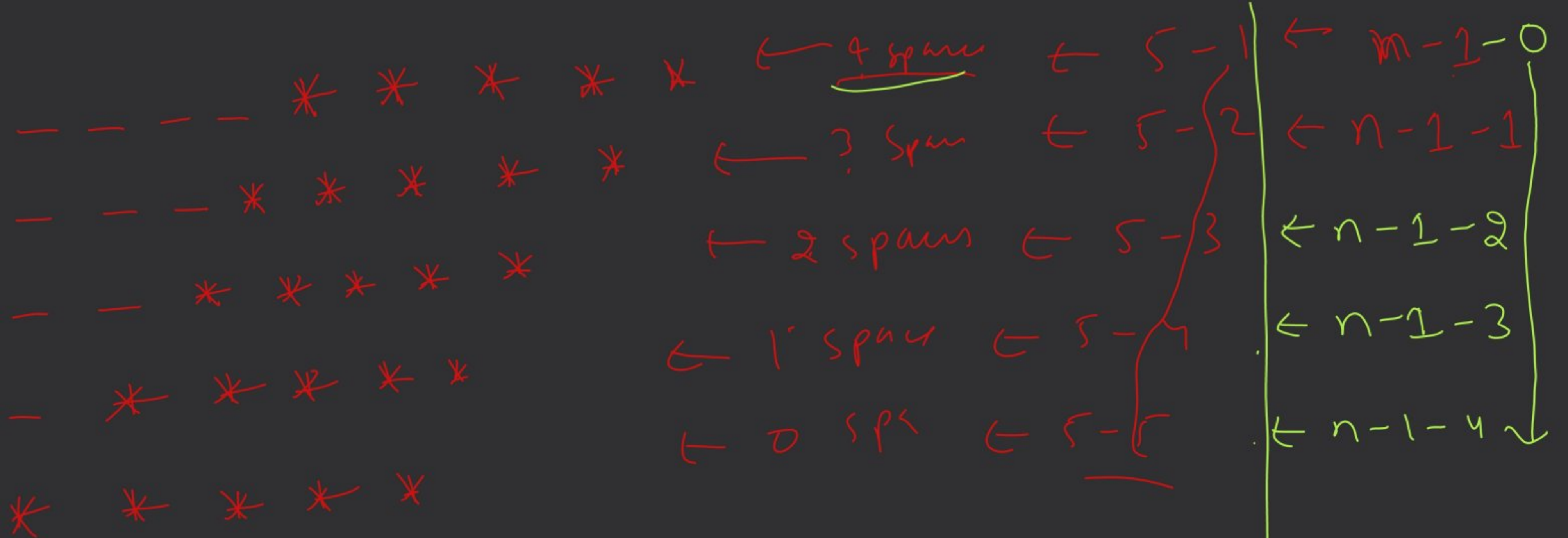
	00	10	11	20	21	4	-	-	-	}}
Sum at index	↓ 0	↓ 1	↓ 2	2	3	4	-	-	-	



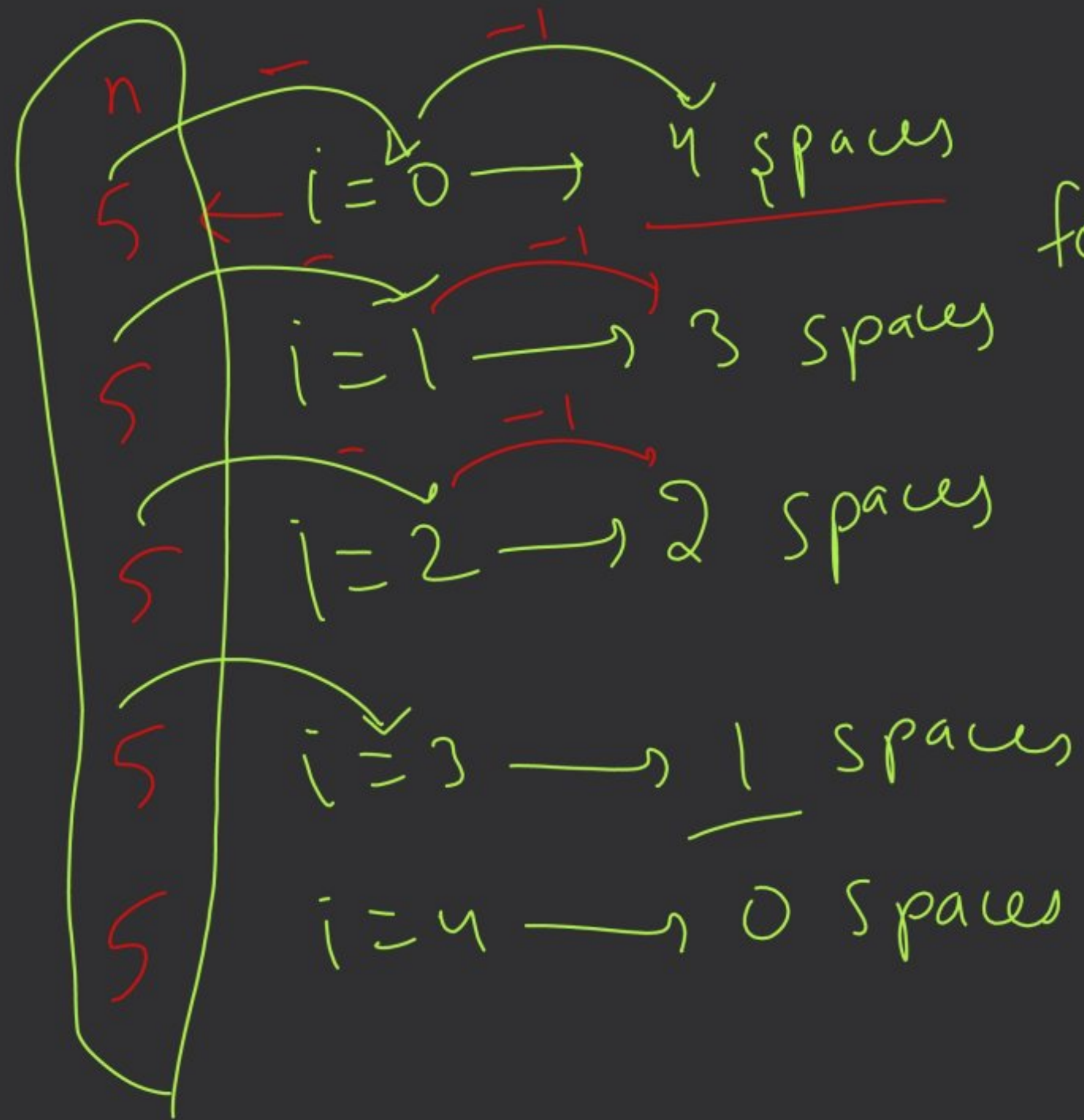
```

for (int i=0; i<n; i++) {
    for (int j=0; j<=i; j++) {
        if ((i+j)%2 == 0) {
            cout<<"1 ";
        } else {
            cout<<"0 ";
        }
        cout<<endl;
    }
}

```

$n=5$



for ($i=0$; $i < 5$; $i++$) { $i=0 \rightarrow 4$ }

// spaces

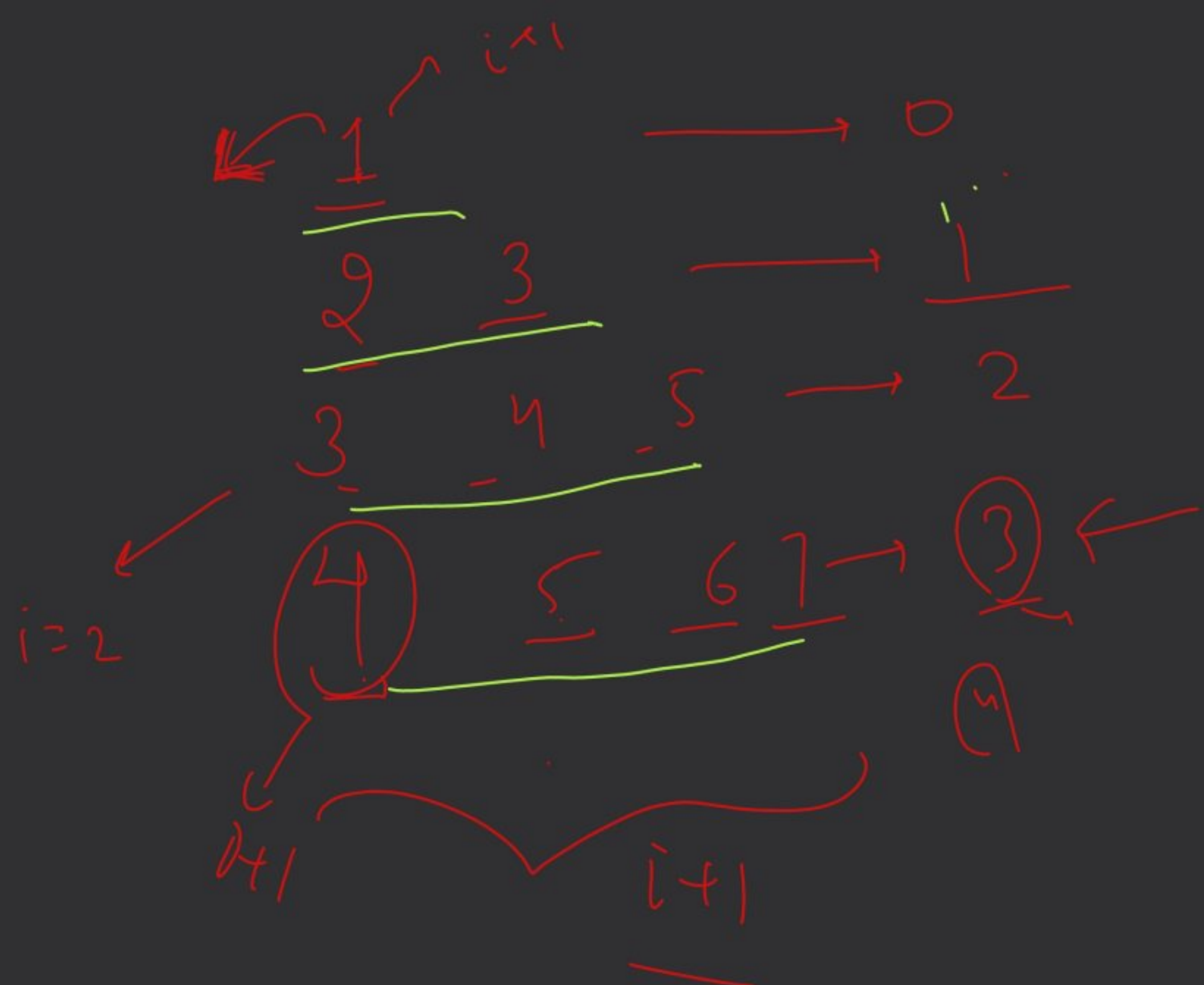
// stars

(0, 1, 2, 3, 4)

$i < r$

$\text{stars} = 0 \rightarrow 4$
 $\text{cout}(" * ");$

$\text{spaces} = n - i - 1$



```

j < 1
i = 0, 1, 2, 3
j < 4
min = 0
for (j = 0; j < i+1; j++) {
    if (arr[j] < min) {
        min = arr[j];
    }
}
cout << min;

```

Handwritten code snippet showing the logic for finding the minimum element in an array using nested loops.

Outer loop index i (values: 0, 1, 2, 3):

- $j < 1$
- $j < 4$

Inner loop index j (values: 0, 1, 2, 3):

- $j = 0 \rightarrow j < i+1; j++$

Minimum value is updated: $min = i+1$.

Output: $cout << min$.

for (i=0 → 3) no. of element

1
3
5
7

for (int i=0 → i<4) {
int num = i+1;

for (int j=0; j<(2*i+1))

System.out.print

(num + " ");

if (j<i) num++;

else num--;

0 ← 1
1 ← 2 3 → 2
2 ← 3 4 5 →
3 ← 4 5 6 →

j=4
j=5
j=6

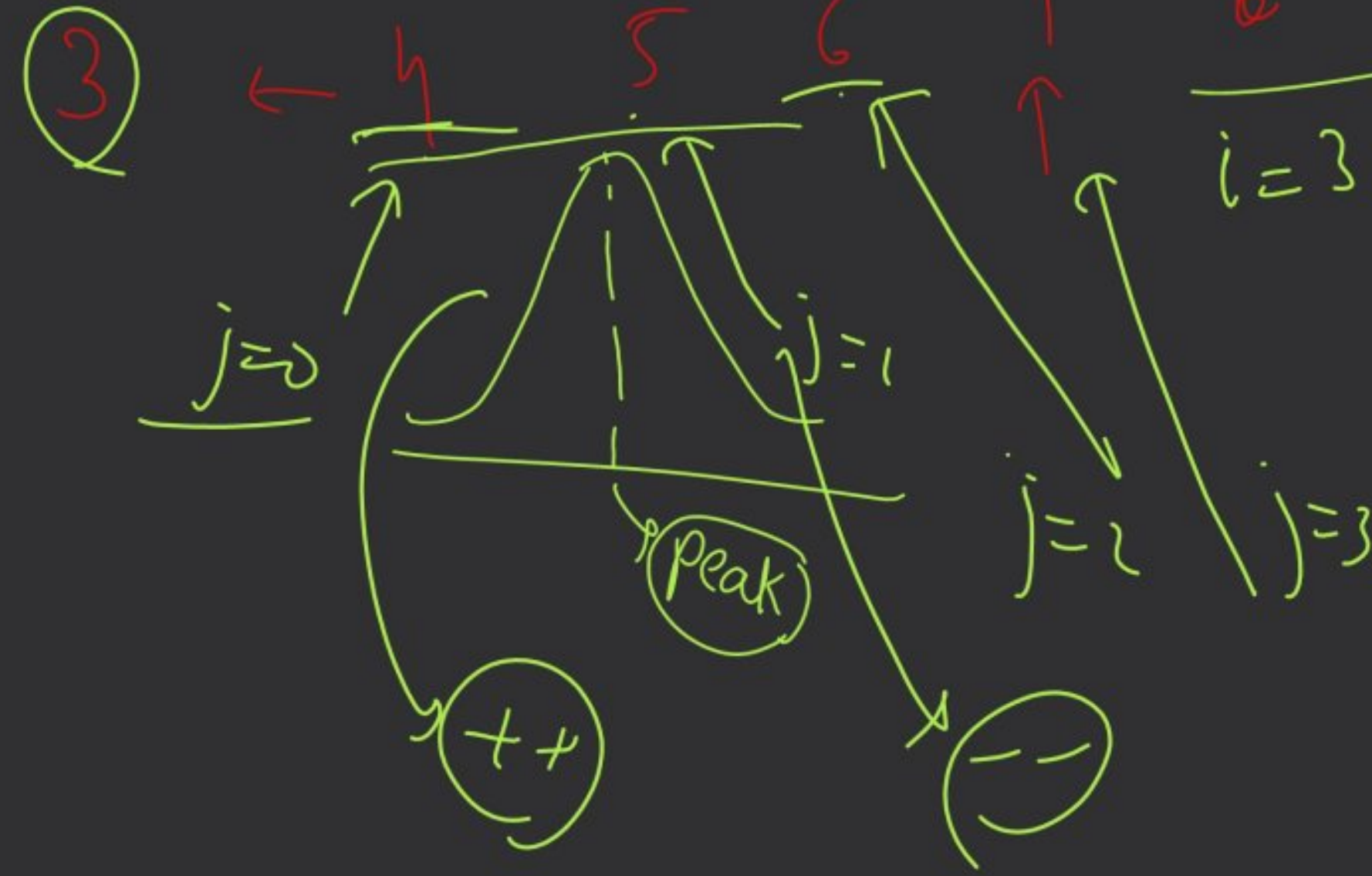
i=3, num=4 j<(2*i+1=7)

0<3 → num=5

1<3 → num=6

2<3 → num=7

3<3 → num=6



$\frac{1}{2}$ 3 2 ←
 $\frac{3}{4}$ 5 4 3
 $\frac{4}{5}$ 6 7 6 5 4

1
 3
 5
 7

$\rightarrow 2 * n + 1$
 $\rightarrow 2 * i + 1$

(Rows = 4) $\rightarrow i = 0, 1, 2, 3$

$i = 0, j = 0 \rightarrow j < 1$ ($j = 0$) num = 0 + 1 = 1

$i = 1, j = 0 \rightarrow j < 3$ ($j = 0, 1, 2$) num = 2

is ($\underline{0 < 1}$) $\rightarrow$ num = 3

$\underline{1 < 1} \rightarrow$ (num = 2)

Output Screen

1
 2 3 2

4 5 6 7 6 5 4

$i=3$, $num=i+1=4$, $j=0 \rightarrow j < 7$ ($\downarrow \downarrow \downarrow \downarrow$)
(~~0~~, ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~5~~, ~~6~~)

$0 < 3$ ✓ $num=5$

$1 < 3$ ✓ $num=6$

$2 < 3$ ✓ $num=7$

$3 < 3$ ✗ $num=6$

$4 < 3$ ✗ $num=5$

$5 < 3$ ✗ $num=4$

$6 < 3$ ✗ $num=3$

($j=1$)

Output

4 5 6 7 6 5 4

H.W. Assignment

→ $n=7$
↓
Do the Dry Run for

$n = 1, 4, 6, 7$
Row → ^{max} 0, 3, 5, 6

| Marks $\geq 90 \rightarrow$ Grade A

Marks $\geq 75 \rightarrow$ Grade B

Marks $\geq 60 \rightarrow$ Grade C

else (fails)

$$2 + 3 = 5$$

|| |||

SolidRhombus.java > SolidRhombus > main(String[])

```
1 public class SolidRhombus {
    Run | Debug
2     public static void main(String[] args) {
3
4         int n = 5; // number of rows
5
6         for (int i = 0; i < n; i++) {
7
8             // spaces
9             for (int space = 0; space < n - i - 1; space++) {
10                 System.out.print(" ");
11             }
12
13             // stars
14             for (int star = 0; star < n; star++) {
15                 System.out.print("*");
16             }
17
18             System.out.println();
19         }
20     }
21 }
```

0 1 2 3 4

(spaces) (stars)

[constant = 5]

BinaryTriangle.java > BinaryTriangle > main(String[])

```
1 //Binary Triangle Pattern
2
3
4 public class BinaryTriangle {
5     Run | Debug
6     public static void main(String[] args) {
7
8         int n = 4; // number of rows
9
10        for (int i = 0; i < n; i++) {
11            for (int j = 0; j <= i; j++) {
12                if ((i + j) % 2 == 0) {
13                    System.out.print(s: "1 ");
14                } else {
15                    System.out.print(s: "0 ");
16                }
17            }
18            System.out.println();
19        }
20    }
21 }
22 }
```

0, 1, 2, 3

n=4

i=0, j=0

1

i=1, j=0, 1